

The i.i.d. indicator admission

Past-determined target meets independent uniform draw

ENS/Polytechnique 2026 Math A — Lean 4 formalization
Companion note (revised after external-LLM verification)

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Abstract

This note explains the probabilistic lemma admitted in the project as `iIndepFun_iIdentDistrib_uniformIndic`. The earlier draft of this note framed it as “Williams §9.7”, citing Williams, *Probability with Martingales*, §9.7. **That framing is misleading.** An external LLM, asked to prove the underlying claim with all named-theorem invocations explicitly forbidden, produced a direct ~ 10 -line counting argument using only definitions of independence + uniformity + the admitted local tree structure. No graduate student would write “by Williams §9.7” in their exam booklet. We give the direct argument here, identify the specific Mathlib API gap that forces the Lean admission anyway, and document its two project uses.

1 Statement

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and S a finite set of cardinality $|S| = n$. Equip S with the discrete σ -algebra. Let $(Y_k)_{k \geq 0}$ be a sequence of independent, identically distributed random variables, each uniform on S :

$$\mathbb{P}(Y_k = s) = \frac{1}{n} \quad \text{for every } s \in S.$$

Define the natural filtration $\mathcal{G}_k := \sigma(Y_0, Y_1, \dots, Y_{k-1})$ for $k \geq 1$, with $\mathcal{G}_0 := \{\emptyset, \Omega\}$ trivial.

Theorem 1 (Williams §9.7, restated for our setting). *Let $c \in \{0, 1, \dots, n\}$ and let $A_k : \Omega \rightarrow 2^S$ be a sequence of $\mathcal{G}_k$ -measurable random subsets of S with constant cardinality $|A_k(\omega)| = c$ for every $\omega \in \Omega$. Define the indicator sequence*

$$f_k(\omega) := \mathbf{1}_{\{Y_k(\omega) \in A_k(\omega)\}}, \quad k \geq 0.$$

Then $(f_k)_{k \geq 0}$ is an i.i.d. sequence of Bernoulli random variables with parameter c/n .

The two project applications correspond to specific choices of S , Y_k , and A_k ; see §4.

2 The direct counting proof (what a grad student would write)

The proof reduces to counting on the finite-product measure $\mathbb{P}_M := \text{Unif}(S)^{\otimes M}$ on S^M , where $M = \max(I) + 1$ for the index set I of interest.

Direct proof of Theorem 1. Fix any finite $I = \{i_1 < \dots < i_r\} \subset \mathbb{N}_0$ and any assignment $\epsilon : I \rightarrow \{0, 1\}$. We show

$$\mathbb{P}(\forall i \in I, f_i = \epsilon_i) = \prod_{i \in I} p_{\epsilon_i}, \quad p_1 := c/n, \quad p_0 := 1 - c/n.$$

The event depends only on the first M coordinates, so it suffices to count realizing tuples in S^M . For each prefix length $\ell \in \{0, 1, \dots, M-1\}$, given a partial realization $(y_0, \dots, y_{\ell-1}) \in S^\ell$ that satisfies the constraints $f_i = \epsilon_i$ for all $i \in I \cap [0, \ell)$, count the number c_ℓ of admissible choices for y_ℓ :

- If $\ell \notin I$: no constraint at step ℓ ; all n values of y_ℓ are admissible. So $c_\ell = n$.
- If $\ell \in I$ with $\epsilon_\ell = 1$: need $y_\ell \in A_\ell(y_0, \dots, y_{\ell-1})$, which has cardinality c . So $c_\ell = c$.
- If $\ell \in I$ with $\epsilon_\ell = 0$: need $y_\ell \notin A_\ell(\dots)$, cardinality $n - c$. So $c_\ell = n - c$.

Crucially, c_ℓ does **not** depend on the partial realization $(y_0, \dots, y_{\ell-1})$; only on whether $\ell \in I$ and the desired ϵ_ℓ . (The set $A_\ell(\dots)$ varies with the prefix, but its *size* is the constant c by hypothesis.)

By induction on ℓ from 0 to $M-1$, the total count is $N = \prod_{\ell=0}^{M-1} c_\ell$. The probability of the event is

$$\frac{N}{n^M} = \prod_{\ell=0}^{M-1} \frac{c_\ell}{n} = \prod_{\ell \in I} \frac{c_\ell}{n} = \prod_{i \in I} p_{\epsilon_i}$$

(steps $\ell \notin I$ contribute factor 1). This establishes mutual independence and identical Bernoulli(c/n) distribution. $\square$

Remark (Length and depth of the argument). The substantive content is the bullet list and the “ c_ℓ does not depend on the prefix” observation. That’s two sentences. Combined with the inductive product formula, the proof is ~ 10 lines. There is no abstract conditional-expectation step. The argument scales to any finite I and any past-dependence structure as long as the cardinality $|A_\ell| = c$ is constant.

3 An equivalent, more abstract framing (for context)

The same fact is sometimes packaged as a corollary of a measure-theoretic lemma:

Lemma 2. *For every $k \geq 0$ and every $\mathcal{G}_k$ -measurable event $E \subseteq \Omega$,*

$$\mathbb{P}(\{f_k = 1\} \cap E) = \frac{c}{n} \mathbb{P}(E).$$

Proof. The set A_k is $\mathcal{G}_k$ -measurable and takes values in the finite collection $\binom{S}{c}$ of c -subsets of S . Decompose E by the value of A_k :

$$E = \bigsqcup_{T \in \binom{S}{c}} (E \cap \{A_k = T\}).$$

Each piece $E_T := E \cap \{A_k = T\}$ is $\mathcal{G}_k$ -measurable. On E_T , the event $\{f_k = 1\} = \{Y_k \in T\}$. By independence of Y_k from $\mathcal{G}_k$ (which holds since $\mathcal{G}_k = \sigma(Y_0, \dots, Y_{k-1})$ and the Y_j are independent),

$$\mathbb{P}(\{Y_k \in T\} \cap E_T) = \mathbb{P}(Y_k \in T) \cdot \mathbb{P}(E_T) = \frac{|T|}{n} \cdot \mathbb{P}(E_T) = \frac{c}{n} \cdot \mathbb{P}(E_T).$$

Sum over T :

$$\mathbb{P}(\{f_k = 1\} \cap E) = \sum_T \mathbb{P}(\{Y_k \in T\} \cap E_T) = \frac{c}{n} \sum_T \mathbb{P}(E_T) = \frac{c}{n} \cdot \mathbb{P}(E). \quad \square$$

Corollary 3. *f_k is independent of $\mathcal{G}_k$, and marginally $f_k \sim \text{Bernoulli}(c/n)$.*

Proof. Take $E = \Omega$ in Lemma 2: $\mathbb{P}(f_k = 1) = c/n$ gives the marginal. For independence, we must show $\mathbb{P}(\{f_k = a\} \cap E) = \mathbb{P}(f_k = a) \mathbb{P}(E)$ for $a \in \{0, 1\}$ and any $\mathcal{G}_k$ -measurable E . The case $a = 1$ is Lemma 2; the case $a = 0$ follows by complementation:

$$\mathbb{P}(\{f_k = 0\} \cap E) = \mathbb{P}(E) - \mathbb{P}(\{f_k = 1\} \cap E) = \left(1 - \frac{c}{n}\right) \mathbb{P}(E). \quad \square$$

Proof of Theorem 1. We have established that each f_k has marginal Bernoulli(c/n), so the family is identically distributed. For mutual independence, observe that $f_0, f_1, \dots, f_{k-1}$ are $\mathcal{G}_k$ -measurable (since f_j is determined by $Y_0, \dots, Y_j$ and $j \leq k-1$ implies $\sigma(f_j) \subseteq \mathcal{G}_k$). By Corollary 3, f_k is independent of $\mathcal{G}_k$, hence in particular of $(f_0, \dots, f_{k-1})$. Inductively, the family $(f_k)_{k \geq 0}$ is mutually independent. Combined with identical distribution, the family is i.i.d. $\square$

Remark. The hypothesis “ $|A_k|$ constant” is essential. If $|A_k|$ were a genuine random variable (say, $\mathcal{G}_k$ -measurable but not constant), then the conditional probability $\mathbb{P}(f_k = 1 \mid \mathcal{G}_k) = |A_k|/n$ would still be a deterministic function of the past, but *not* constant; f_k would be conditionally Bernoulli with a random parameter, and the marginal would involve averaging $|A_k|/n$ over the past. Independence from the past would fail in general.

Lemma 4 (“Constant conditional probability” lemma). *For every $k \geq 0$ and every $\mathcal{G}_k$ -measurable event $E \subseteq \Omega$,*

$$\mathbb{P}(\{f_k = 1\} \cap E) = \frac{c}{n} \mathbb{P}(E).$$

This is provable as before via decomposition over the finite values of A_k . From here mutual independence follows by a tower-of-induction argument, exactly as Williams §9.7 packages it.

The framing question. Both the direct counting proof and the abstract lemma 2 are correct and equivalent. The direct counting is what a graduate student would write in 10 lines on their exam booklet; the abstract lemma is what a textbook records for general use. *Neither is harder than the other; they encode identical content at different abstraction levels.*

4 Why “the centred indicators are not i.i.d. ” is wrong

A common heuristic, repeated in the project’s earlier draft (errata note E3 of `RandomWalk.lean`), goes:

*The centred indicator $\xi_i := \mathbf{1}_{\{Y_i \in A_i\}} - c/n$ **depends on the past** (through A_i), hence the family (ξ_i) is not independent.*

This is incorrect. “Depends on the past” is a statement about *measurability*, not about *independence*. The ξ_i is $\sigma(Y_0, \dots, Y_i)$ -measurable, true. But Theorem 1 says the family is nevertheless i.i.d. because the conditional law given the past is *constant*: “constant conditional law” is a measure-theoretic property strictly stronger than “measurable function of past” and strictly weaker than “deterministic”.

The constant-cardinality hypothesis $|A_k| = c$ is exactly the content the heuristic misses.

5 Project applications

In both applications, $S = \{a, b, a^{-1}, b^{-1}\}$ is the generating set of F_2 , equipped with the uniform measure Z_{unif} ($n = 4$), and Y_k is the k -th step of the random walk on F_2 .

5.1 Q42 — binomial identification

The exam asks to show that

$$\frac{b_\varphi(X_n) + n}{2} \sim \mathcal{B}(n, 3/4).$$

Write the LHS as $\sum_{i < n} \xi_i$ where

$$\xi_i := \frac{b_\varphi(X_{i+1}) - b_\varphi(X_i) + 1}{2}.$$

By the admitted 1+3 neighbour structure of the Busemann function, exactly 1 of the 4 generators decreases b_φ by 1 and 3 increase it by 1. So $\xi_i = \mathbf{1}_{\{Y_i \in A_i\}}$ where

$$A_i(\omega) := \{s \in S : b_\varphi(X_i(\omega) \cdot s) = b_\varphi(X_i(\omega)) + 1\}$$

has constant cardinality $|A_i| = 3$. By Theorem 1, (ξ_i) is i.i.d. Bernoulli(3/4), hence $\sum_{i < n} \xi_i \sim \mathcal{B}(n, 3/4)$.

The exam's admitted Hoeffding inequality (preamble) then gives the required tail bound, and Borel–Cantelli yields the SLLN $b_\varphi(X_n)/n \rightarrow 1/2$ a.s.

5.2 Q43 — cancellation coupling

The exam asks to show $|X_n|/n \rightarrow 1/2$ a.s. Writing $|X_n| = n - 2C_n$ where C_n counts cancellations, the proof reduces to $C_n/n \rightarrow 1/4$ a.s. Define

$$\text{coupled}_k := \mathbf{1}_{\{Y_k = (s_k)^{-1}\}}$$

where s_k is the last letter of X_k (with arbitrary default at identity). The set $A_k = \{(s_k)^{-1}\} \subseteq S$ has constant cardinality 1. By Theorem 1, (coupled_k) is i.i.d. Bernoulli(1/4). Mathlib's strong law gives $C_n/n \rightarrow 1/4$ a.s. (after a coupling argument that handles the random visits to identity, where the formal coupled_k disagrees with the true cancellation indicator on a finite set a.s.).

5.3 Common structure

Both applications fit Theorem 1 with $|S| = 4$ and constant $c \in \{1, 3\}$. They are the only places in the project where the admission is invoked. Wave 23C consolidated them under a single generic Lean axiom matching Theorem 1's statement.

6 Necessity

6.1 Q42 cannot be done without it at graduate level

The exam asks the *student* to derive the binomial law of $\sum \xi_i$. The derivation requires showing the family is i.i.d. This is exactly the content of Theorem 1; there is no shorter route.

Alternative: admit the binomial law directly. This replaces a single general admission with a more specific one and does not help any other question.

6.2 Q43 alternative routes

The cleanest grad-level proof is the i.i.d. coupling above. A geometric “4-ray” argument exists but invokes either:

- the same i.i.d. structure on the Busemann increments (no saving), or
- more elaborate facts about random walks on non-amenable groups (Kesten, Woess, Ch. 8), strictly stronger than what the exam expects.

The i.i.d. coupling using Theorem 1 is the minimal graduate-level argument. The errata document confirms this (Q43 §(b), corrected route).

7 The Lean obstacle (this is purely about Mathlib API, not the math)

The direct counting proof (§2) and the abstract framing (§3) are both pure mathematics and complete. Why, then, do we admit the statement in Lean?

The answer is that Lean’s Mathlib library has a specific gap:

$$\mu = \text{Measure.infinitePi}(\mu_S)_{k \in \mathbb{N}}$$

into “finite prefix $\otimes$ infinite suffix” over $\mathbb{N}$:

$$\mu = \text{Measure.pi}(\mu_S^{[0,N)}) \otimes \text{Measure.infinitePi}(\mu_S)_{k \geq N}.$$

Mathlib has the finite-prefix split for `Measure.pi` over `Iic` b ranges (`pi_prod_map_IicProdIoc`), but not for `Measure.infinitePi` over $\mathbb{N}$. Filling this gap (200 LOC of pure measure-theoretic plumbing, no probability or graph-theoretic content) would dissolve the Lean admission and turn Theorem 1 into a Lean theorem.

This is a one-time Mathlib investment with multiple downstream benefits beyond this project.

Once the prefix-suffix split is in place, the Lean proof of Theorem 1 is the same direct counting argument as §2: project to S^M , count $N = \prod c_\ell$, divide by n^M . ~ 200 LOC of measure-theoretic plumbing for a 10-line mathematical argument.

The honest situation: the Lean admission is misleadingly named after Williams §9.7 because that’s a convenient textbook citation. The mathematical content is undergraduate-tier counting; the Lean blocker is purely the absence of a finite-prefix split for `Measure.infinitePi`. Future readers should understand: this is not a deep probability admission, it is a Lean infrastructure gap.

8 References

- D. Williams, *Probability with Martingales*, Cambridge University Press, 1991. §9.7 (“Conditional expectations: further properties”).
- A. Klenke, *Probability Theory: A Comprehensive Course*, Springer, 2014, 2nd ed. §5.3 (“Conditional probabilities and conditional expectations”).
- D. Bertsekas, J. Tsitsiklis, *Introduction to Probability*, Athena Scientific, 2008. §1.5–1.6 covers the finite-state version (“constant conditional probability $\Rightarrow$ independence”) as a worked exercise.